

Water Potability Prediction - Project Report

This report summarizes an end-to-end binary classification project that predicts whether water is potable using physicochemical properties.

Dataset

3276 samples, 9 input features and 1 binary target (Potability). Missing values: Sulfate 625 (23.85%), pH 387 (14.77%), Trihalomethanes 134 (5.11%). Median imputation was used.

Preprocessing

Missing values imputed, log1p transformations for skewed features, StandardScaler applied where required, interaction feature Conductivity × Hardness created.

Models

Logistic Regression, Random Forest, Gradient Boosting, SVM (RBF), KNN and Gaussian Naive Bayes were trained and compared.

Evaluation

Metrics: Accuracy, Precision, Recall, F1-score and ROC-AUC.

Best Results

Random Forest achieved the highest ROC-AUC (0.6395). Gradient Boosting achieved the highest accuracy (0.6540). SVM obtained the best F1-score (0.4990).

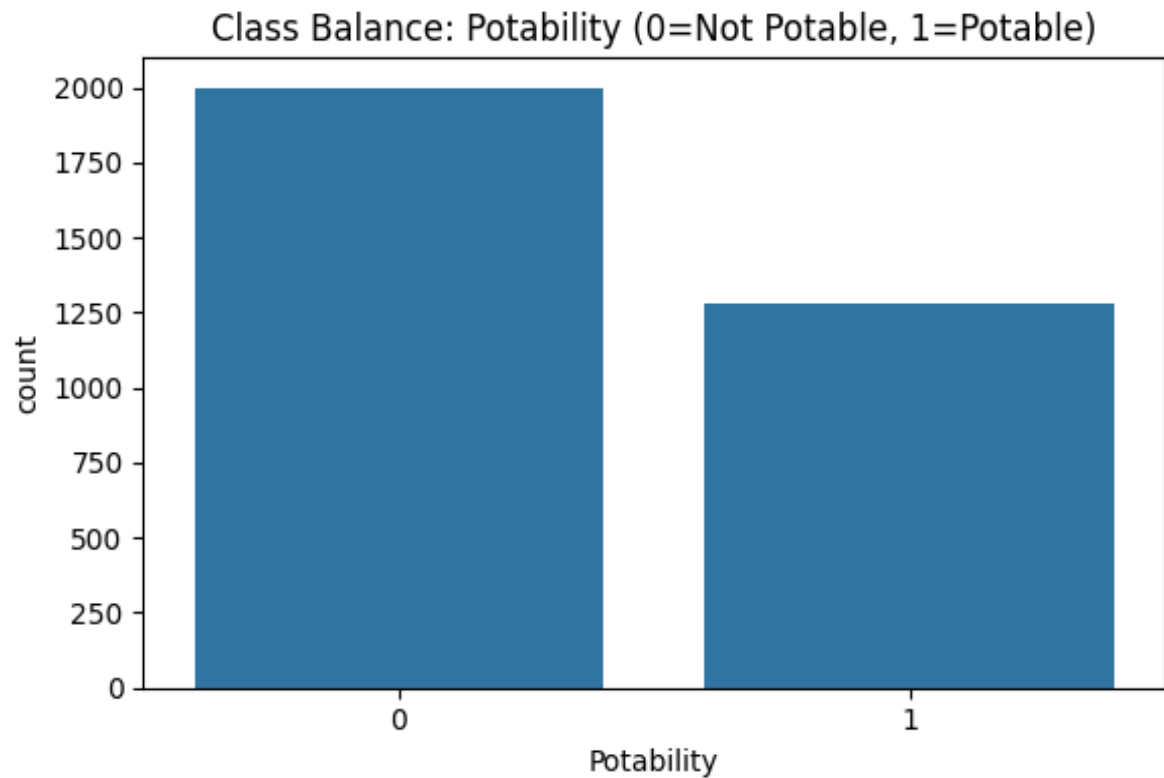
Observations

Permutation importance identified Sulfate and pH as the strongest predictors. The dataset is moderately imbalanced (more non-potable than potable samples), making ROC-AUC and F1-score important evaluation metrics.

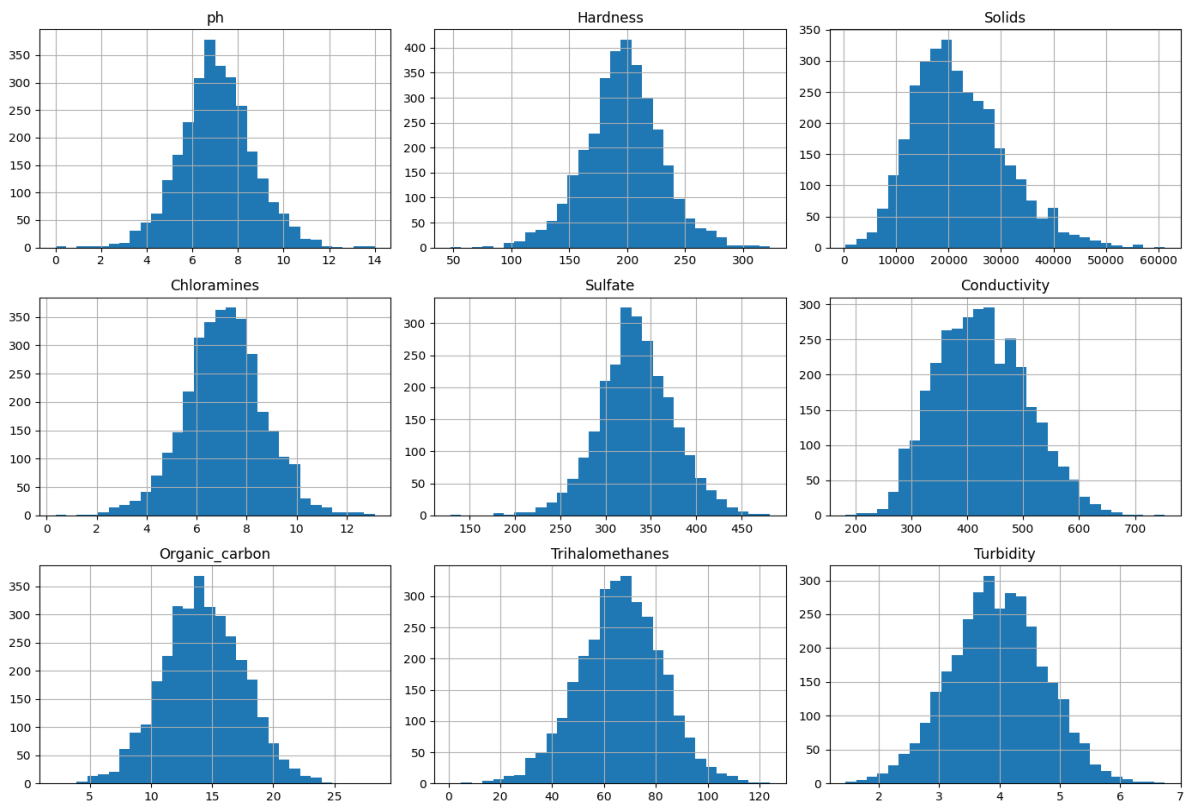
Conclusion

The workflow demonstrates preprocessing, feature engineering, model comparison and interpretation. Future work includes hyperparameter tuning, SMOTE/class balancing and XGBoost/LightGBM.

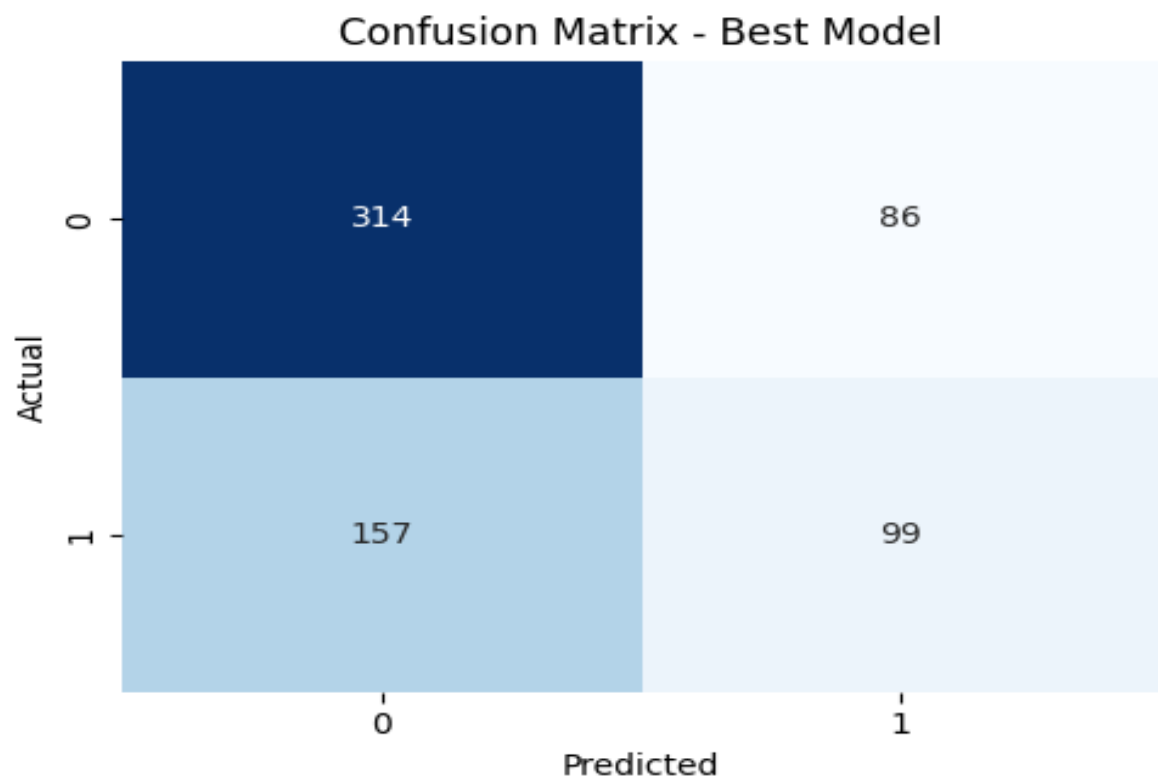
Feature Distributions & EDA



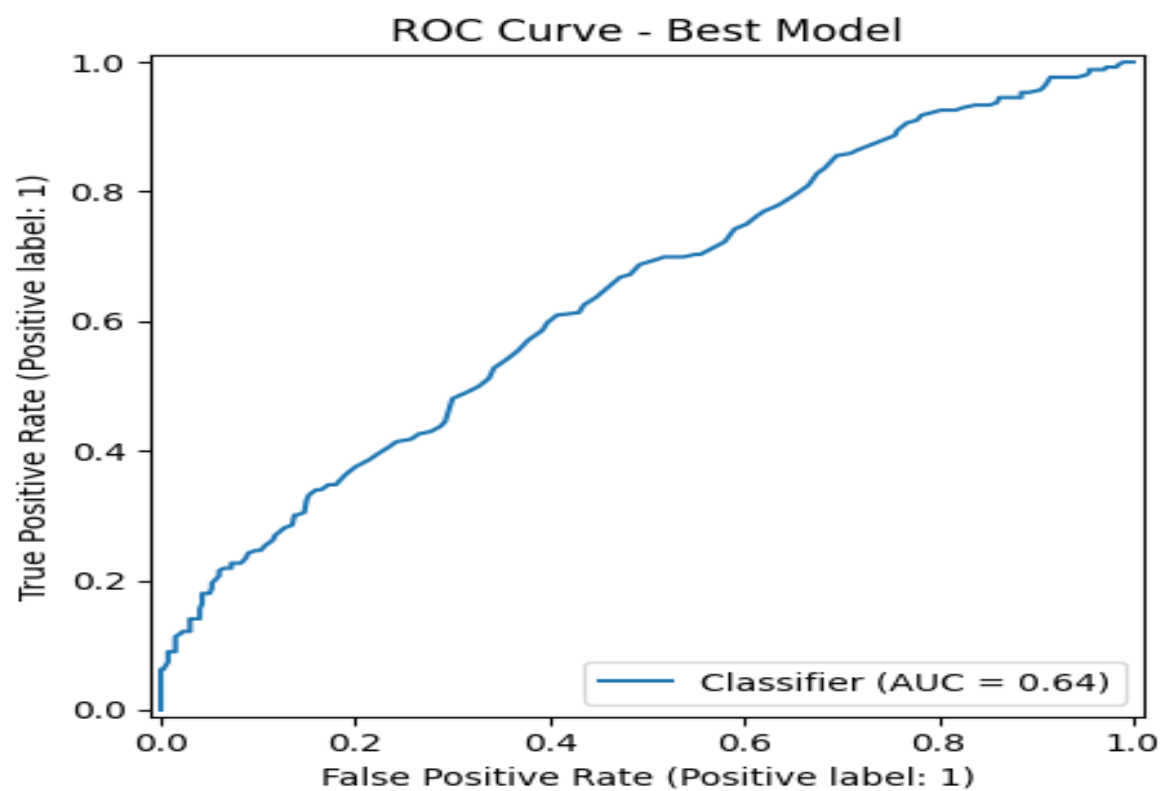
Class Balance



Confusion Matrix



ROC Curve



Permutation Feature Importance

